Layers of software engineering

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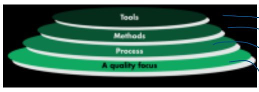
Definitions



LOL no idea why this text
is so blurry srry

TOOLS
Provide (semi?)automated support for the process and methods
When tools are integrated so that information by one tool can be used by another, a system for the support of software development is established
- Computer-aided Software Engineering

METHODS
Provide the technical how-to's for building software
Encompasses a broad array of tasks
- Communication
- Requirements analysis
- Design modelling
- Program construction
- Testing
- Support

These methods rely on a set of basic principles that govern each area of the technology
Includes modelling activities and techniques

PROCESS
The software engineering process holds the technology layers together and enables rational and timely development of softrware
Process defines a framework that must be established for effective delivery of softwarwe engineering technology.

The process forms the basis for management control of software projects and establishes the context in which technical methods are applied, work products are produced, milestones are established, quality is ensured, change is properly managed

The process framework encompasses **framework activities** that are applicable to **ALL** projects

Communication
Planning
Modelling
Construction
Deployment

Communication

Before any technical work can start, it's important for stakeholders to communicate and collaborate with the customer and other stakeholders.

The intent is to understand stakeholders' objectives for the project and to gather requirements that help define software features and functions

Planning

Any complicated journey can be simplified if a map exists
A software project is a complicated journey and the planning activity creates a map that helps guide the team as it makes the journey.

The software project plan defines the software engineering work by describing the technical tasks to be conducted
- the risks that are likely
- the resources that will be required
- the work products to be produced
- the work schedule

Modelling

You work with models every day no matter what type of eng you are

You create a *sketch* of a thing so that you'll understand the big picture

- What it looks like architecturally
- How the constituent parts fit together
- Many other characteristics

You refine the sketch into greater detail in an effort to better understand how you're going to solve the problem

A software engineer does the same thing by creating models to better understand software requirements and the design that will achieve those requirements

Construction

Code generation
- Manual or automated

Testing that is required to discover errors in code

Deployment

The software is delivered to a customer who evaluates the product and provides feedback
Either complete or as partially completed increments

The process framework also has **umbrella** activities- - applicable across the entire software PROCESS

Software project tracking and control	Risk management	Software Quality Assurance	Technical reviews	Measurement

Software configuration management	Reusability management	Work product preparation and production

A QUALITY FOCUS
All engineering approaches must rest on an organizational commitment to quality

Total Quality Management
Six Sigma
And similar philosophies foster a continuous process improvement culture
This culture ultimately leads to the development of increasingly more effective approaches to software engineering
The bedrock that supports software engineering is a quality focus